Mahindra University, Hyderabad
Ecole Centrale School of Engineering
Minor-II

Program: B. Tech **Branch: CSE, AI, ICS, ECM, COM, CM** **Semester: I**
Subject: Database Management Systems (CS3103)

Date: 30.10.2025
Time Duration: 1.5 Hours
Time: 2.00 PM - 3.30 PM
Max. Marks: 30 Marks

Instructions:

1) **Any question attempted using pencil will not be considered for the evaluation.**
2) Ensure your answers are well-structured by labeling each question and sub-question and following the sequence in which they appear in the question paper.
3) If a question appears ambiguous, clearly state your assumption(s) and then proceed to answer or solve it accordingly.
4) If a question involves calculations, show all steps and workings clearly.

Q1. Answer the following questions:

1.1 Given the relation instance R(A, B, C, D)

A	B	C	D
1	2	5	3
1	4	5	7
2	2	6	3
3	4	8	7

For each of the following functional dependencies, state whether it holds or does not hold over the given relation R, and briefly justify your answer. **[3 Marks]**

a) $A \rightarrow C$
b) $BD \rightarrow A$
c) $BC \rightarrow A$

1.2 Consider a relation R (A, B, C, D, E, F) with the following set of functional dependencies: $F = \{A \rightarrow B, AB \rightarrow C, BC \rightarrow D, CD \rightarrow E, DE \rightarrow F, B \rightarrow D\}$
For each of the following functional dependencies, determine whether it is implied by F. If implied, show the closure computation that proves it. If NOT implied, show the closure computation that shows the RHS is not included. **[3 Marks]**

a) $AB \rightarrow EF$
b) $BC \rightarrow DE$
c) $BC \rightarrow AF$

Q2. Consider a relation R (A, B, C, D, E, G) with the following set of functional dependencies:

$F = \{AB \rightarrow C, AC \rightarrow B, AD \rightarrow E, B \rightarrow D, BC \rightarrow A, E \rightarrow G\}$ **[3+4=7 Marks]**

The relation R is decomposed into the following sub-relations:

R1(A, B, C) R2(A, C, D, E) R3(A, D, G)

Using the given information, determine whether the decomposition is:

a) Lossless or lossy, and
b) Dependency preserving or not.

Provide proper reasoning to justify your conclusions.

Q3. Consider a relation R(A, B, C, D, E, F, G) with the following set of functional dependencies:

$F = \{AC \rightarrow D, A \rightarrow BD, D \rightarrow E, F \rightarrow G\}$

a) Find the candidate key(s) of the relation R. **[2 Marks]**
b) Compute the canonical cover (minimal cover) of F. **[2 Marks]**
c) Is the given relation R in 3NF? If not, explain the reason of violation & decompose to be in 3NF. **[3 Marks]**

Q4. Consider the relational database where the primary keys are underlined and highlighted in bold. **[5 Marks]**

Employee(<u>**person-name**</u>, street, city)
Works(<u>**person-name**</u>, <u>**company-name**</u>, salary)
Company(<u>**company-name**</u>, city)
Manages(<u>**person-name**</u>, <u>**manager-name**</u>)

Using the given database write the relational algebra queries for the following questions.

a) Find the name of all employees who works for "SBI Bank".
b) Find the names and city of residence of all employees who works for "SBI Bank".
c) Find the names of all employees in this database who live in the same city as the company for which they work.
d) Find the name of all employees in the database who do not work for "SBI Bank".
e) Assume that the company may be located in several cities. Find all the company name located in every city in which "HDFC Bank" is located.

Q5. Consider the following database, where the primary keys are underlined and highlighted in bold. Write the SQL query for the following questions. **[5 Marks]**

Employee(<u>**Empno**</u>, Ename, Job, Hiredate, Basic_Sal, Comm, Dept_no)

a) List the employee name, Job, and department no. of everyone whose name fall in the alphabetical order between "C" to "L" from employee table. (solve the query using between operator)
b) List all employee who were hired during 1999 from the employee table.
c) Find out the difference between highest and lowest salaries of employees.
d) Change the basic_sal = 3000 where job is Clerk from employee table.
e) List all employees whose name starts with 'A' and ends with 'D' from the employee table.